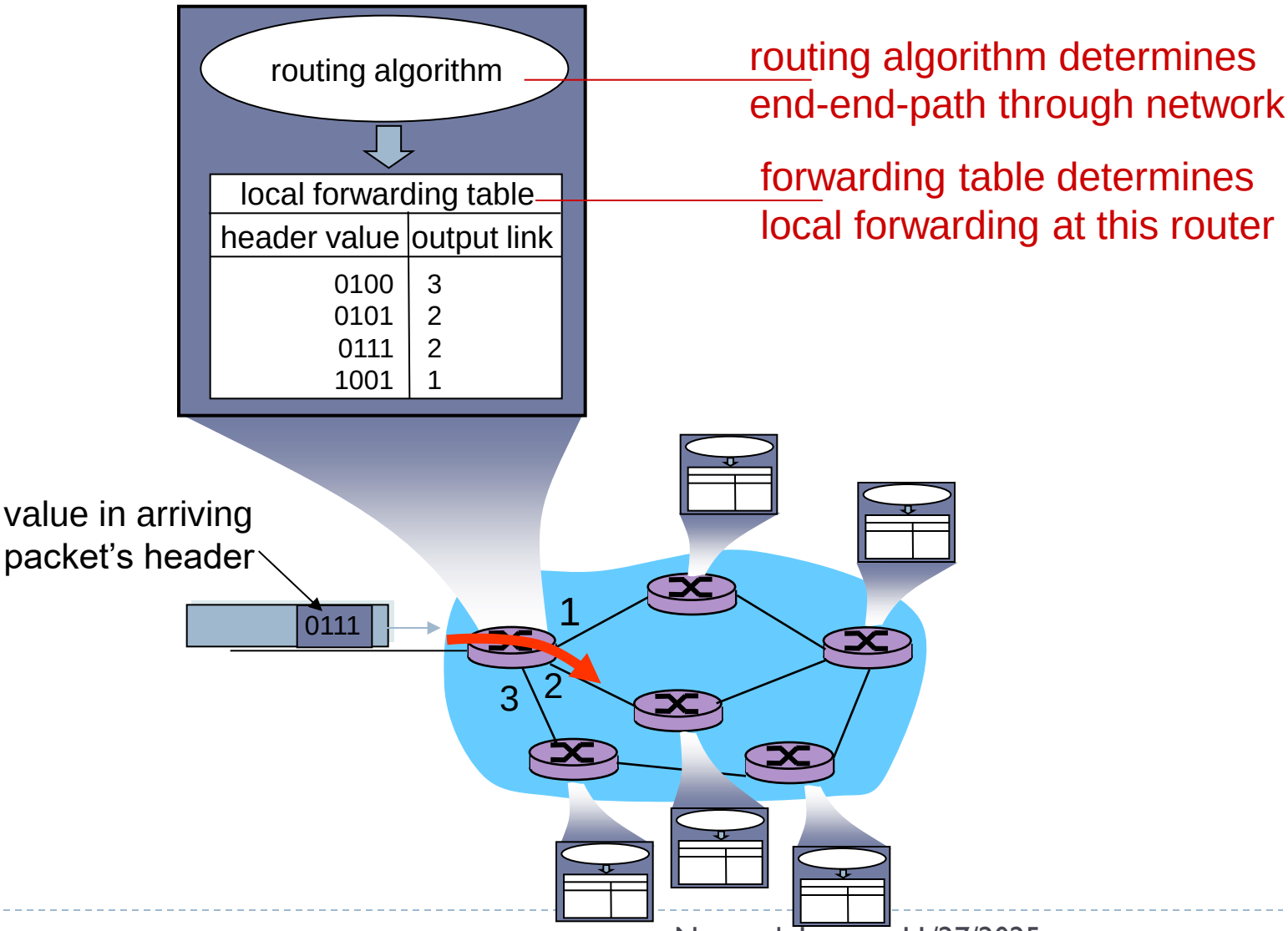
Network layer

Key Function of Network layer

- Role of the network layer is simple – to move packets from a sending host to a receiving host
- Two important network layer functions can be identified

1. **Forwarding**
   - ✓ When a packet arrives at a router's input link, the router must move the packet to the appropriate output link.
2. **Routing**
   - ✓ It's a process of selecting best paths in a network
   - ✓ The network layer must determine the route or path taken by packets they as flow from a sender to a receiver
   - ✓ The algorithms that calculate these paths are referred to as routing algorithms

Interplay between routing and forwarding



Networking Devices

- **Repeater**
  - Repeater is an electronic device that receives a signal and retransmits it.
  - It helps to regenerate or replicate weak or distorted signals.
  - It operates at physical layer.
  - It only repeats the signal without understanding the packets.
  - It does not have physical address on the network.
- **Hub**
  - Hub is a network component that acts as a common connection point for nodes in a network.
  - It contains multiple ports.
  - When a packet arrives at one port, it is copied to all other ports so that all segments of LAN can see all packets.
- **Switch**
  - When a packet arrives at one port, it is copied to only the port that leads to destination node.

Networking Devices

- **Bridge**
  - A network bridge is a network component that creates a single aggregate network from multiple networks.
  - It operates in physical and data link layer.
  - It operates using physical address of machines.
  - The bridge creates function table with machine's address and the segment they belong to.
- **Router**
  - Router is a network device that forwards data packets between networks.
  - A router is connected to two or more data lines from different networks.
  - When a data packet comes in on one of the lines, it reads the address information in the packet to determine the ultimate destination.
  - The information in routing table helps to direct packet to next network.
- **Gateway**
  - Gateway is an inter-networking system capable of joining together two networks that use different base protocols.

Internet Address

Internet Addressing

- Network address is logical address given by the software which can be changed by appropriate configurations.
- A network address always points to host/node/server.
- It is configured on network interface card.
- It is mapped by system with MAC address of machine. Eg: IP addresses
- IP address provides mechanism to differentiate between hosts and network.
- As IP address is assigned in hierarchial manner, a host always resides under a specific network.
- A host in different subnet need a mechanism to locate each other, which is done by DNS.
- DNS is a server which provides layer 3 address of remote host mapped with its domain name.
- When a host acquires layer 3 address of remote host, it forwards packets to its gateway.

IP Address

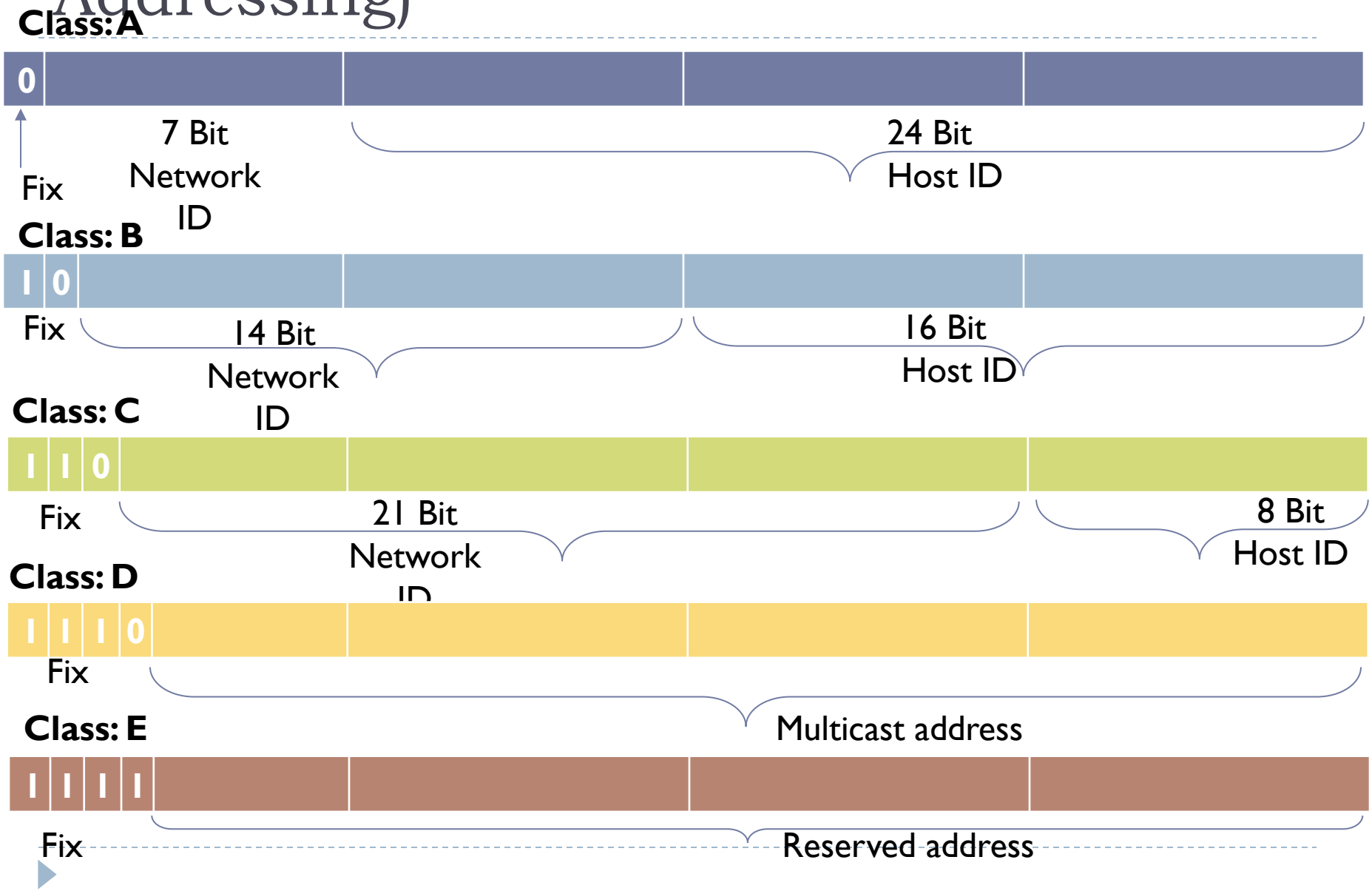
- IP addresses are useful in identifying a specific host in a network.
- IP addresses are 32 bit numbers which are divided into 4 octets. Each octet represents 8 bit binary number. Below is an example of an IP address:

10101100	00010000	11111110	00000001
172	**16**	**254**	**1**

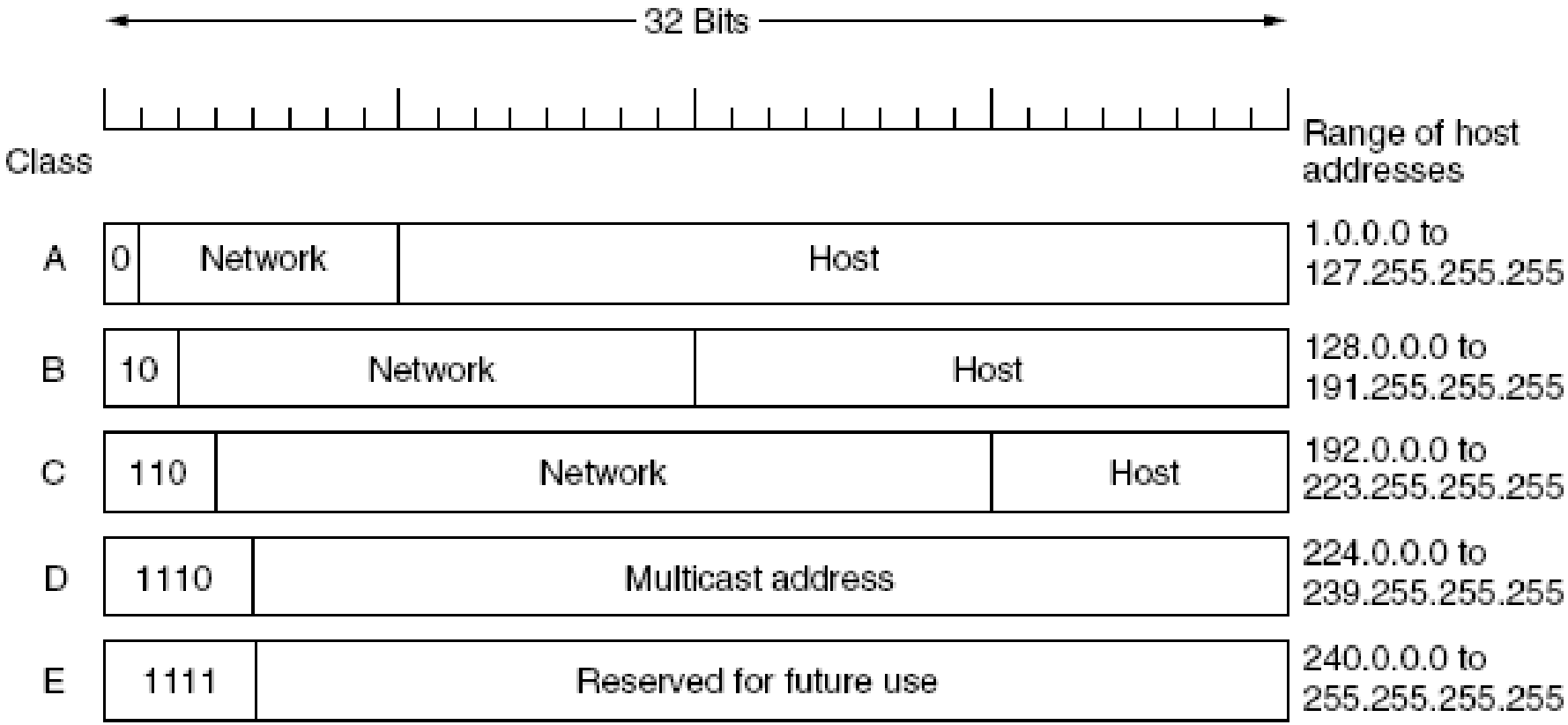
IP addresses are divided into 2 parts:
Network ID & Host ID

<NID> <HID> = IP Address

Classification of IP Addresses (Classful Addressing)



IP Address Formats

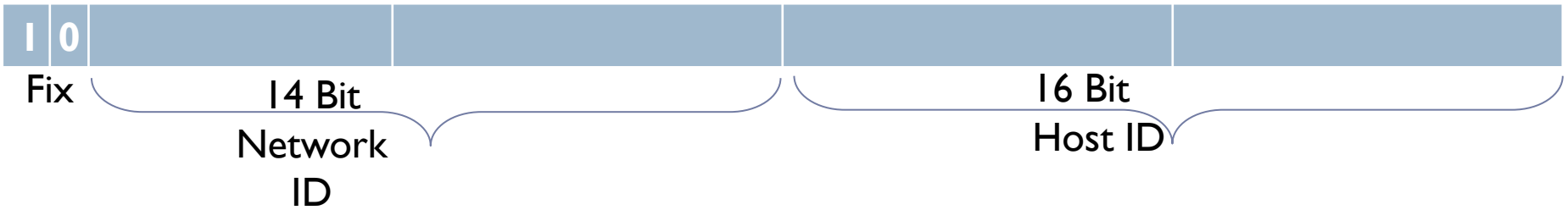


Class A: (0.0.0.0 to 127.255.255.255)

- Only 126 addresses are used for network address.
- All 0's and 1's in Network-ID are dedicated for special IP address. So, total number of IP address in class A can be represented:

0.0.0.0	Special IP Address
00000001.0.0.1 1.0.0.2 1.0.0.3 . . . 126.255.255.254	$2^{24} - 2$ are Host IP
127.255.255.255	Special IP Address – Loopback

Class B: (128.0.0.0 to 191.255.255.255)



- No special network address here. All are usable.

128.0.0.0	Special IP Address
10000001.0.0.1 130.0.0.2 130.0.0.3 . . . 190.255.255.254	$2^{16} - 2$ are Host IP
10111111.255.255.255	Special IP Address – Loopback

Class C: (192.0.0.0 to 223.255.255.255)

192.0.0.0	Special IP Address
11000001.0.0.1 194.0.0.2 194.0.0.3 . . . 222.255.255.254	$2^8 - 2$ are Host IP
11011111.255.255.255	Special IP Address – Loopback

Class D: (224.0.0.0 to 239.255.255.255)

- Very first four bits of the first octet in Class D IP addresses are set to 1110, giving a range of:

11100000 – 11101111
224 – 239

- Class D has IP address rage from 224.0.0.0 to 239.255.255.255.
- Class D is reserved for Multicasting.
- In multicasting data is not destined for a particular host, that is why there is no need to extract host address from the IP address, and Class D does not have any subnet mask.

Class E: (240.0.0.0 to 255.255.255.255)

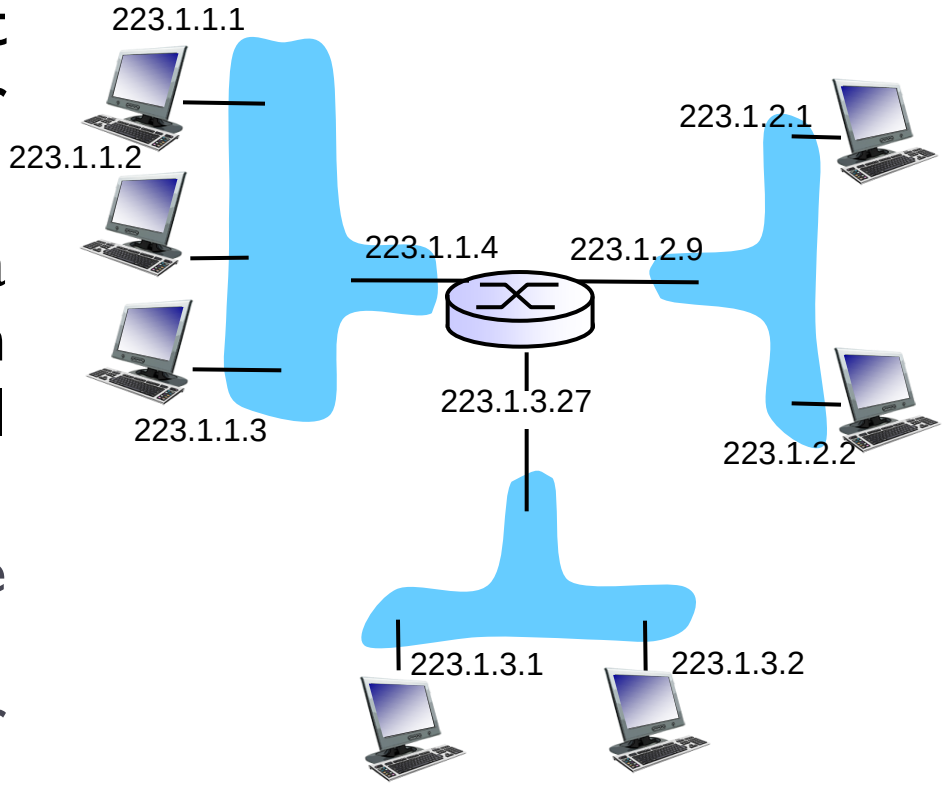
- This IP Class is reserved for experimental purposes only for R&D or Study.
- IP addresses in this class ranges from 240.0.0.0 to 255.255.255.254.
- Like Class D, this class too is not equipped with any subnet mask.

IP Addressing Summary

Class	Leading bits	Size of *network number* bit field	Size of *rest* bit field	Number of networks	Addresses per network	Total addresses in class	Start address	End address	Default subnet mask in dot-decimal notation	CIDR notation
Class A	0	8	24	128 (2^7)	16,777,216 (2^{24})	2,147,483,648 (2^{31})	0.0.0.0	127.255.255.255	255.0.0.0	/8
Class B	10	16	16	16,384 (2^{14})	65,536 (2^{16})	1,073,741,824 (2^{30})	128.0.0.0	191.255.255.255	255.255.0.0	/16
Class C	110	24	8	2,097,152 (2^{21})	256 (2^8)	536,870,912 (2^{29})	192.0.0.0	223.255.255.255	255.255.255.0	/24
Class D (multicast)	1110	not defined	not defined	not defined	not defined	268,435,456 (2^{28})	224.0.0.0	239.255.255.255	not defined	not defined
Class E (reserved)	1111	not defined	not defined	not defined	not defined	268,435,456 (2^{28})	240.0.0.0	255.255.255.255	not defined	not defined

IP Addressing - Example

- **IP address:** It is 32-bit identifier for host, router interface
- **Interface:** It is a connection between host/router and physical link.
  - A router's typically have multiple interfaces
  - A host typically has one or two interfaces
- IP addresses associated with each interface.



223.1.1.1 = $\underbrace{11011111}_{223}$ $\underbrace{00000001}_{1}$ $\underbrace{00000001}_{1}$ $\underbrace{00000001}_{1}$

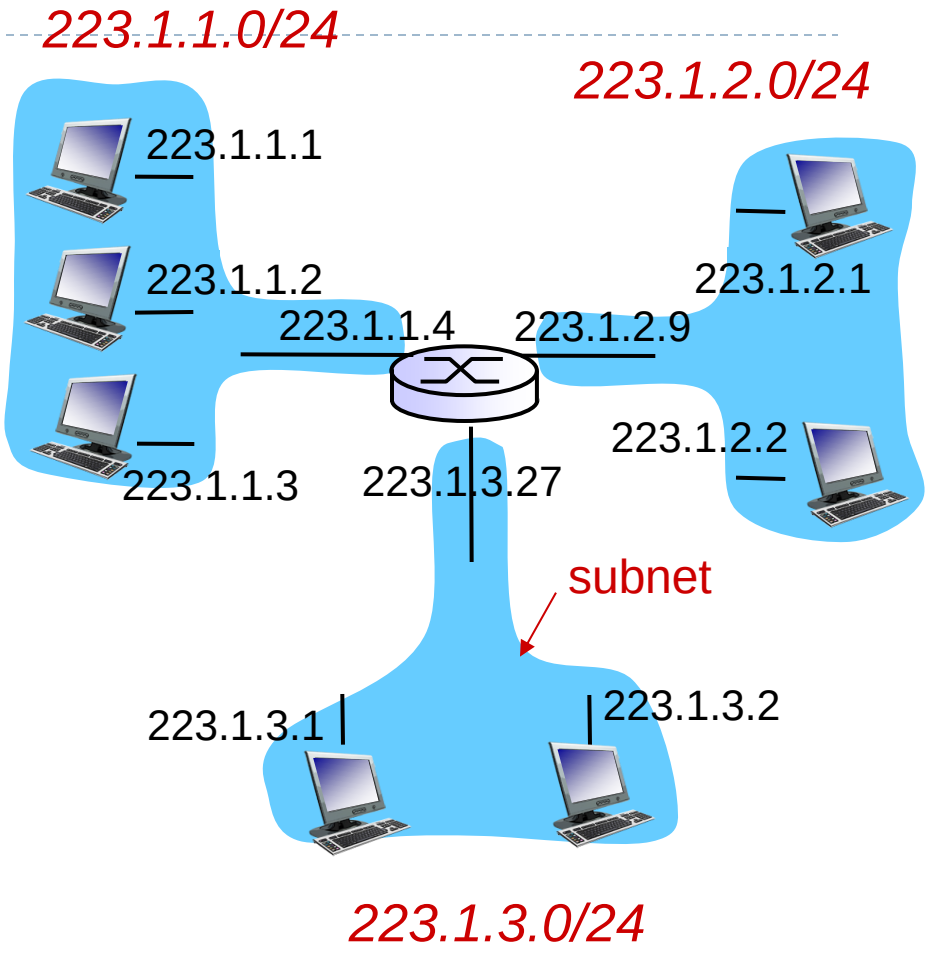
Subnetting

- Subnetting is the process of breaking the class of network into smaller network/subnet.
- It is necessary because a large single network of a class without subnet will not be efficient.
- It splits the host field into subnet creating a three part address.
- The network field remains unchanged which is determined by classful addressing.
- A subnet mask is assigned to determine between subnet and host fields.

Class	*Subnet Mask*
A	*255.0.0.0 or /8*
B	*255.255.0.0 or /16*
C	*255.255.255.0 or /24*

Subnet

- **Subnet**: high order bits defines subnet
- **Host:** low order bits defines host
- To determine the subnets, detach each interface from its host or router.
- Creating islands of isolated networks, with interfaces terminating the end points of the isolated networks.
- Each of these isolated networks is called a **subnet**.



Network consisting of 3 subnets

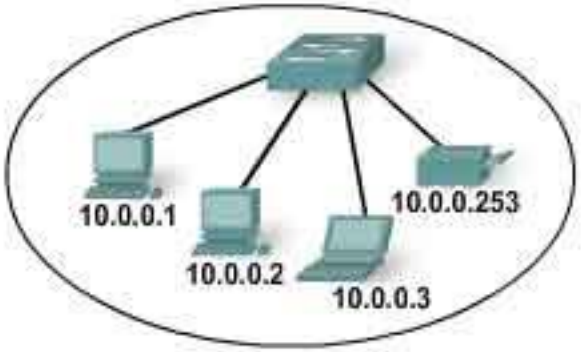
Type of addresses in IPv4 Network

- **Network address -** The address by which we refer to the network.
  - E.g.: 10.0.0.0
- **Broadcast address -** A special address used to send data to all hosts in the network.
  - The broadcast address uses the highest address in the network range.
  - E.g.: 10.0.0.255
- **Host addresses -** The addresses assigned to the end devices in the network.
  - E.g.: 10.0.0.1

Type of addresses – Cont…

Address Types

	Network			Host
Network Address	10	0	0	0
	00001010	00000000	00000000	00000000
Broadcast Address	10	0	0	255
	00001010	00000000	00000000	11111111
Host Address	10	0	0	1
	00001010	00000000	00000000	00000001



Classless Inter-Domain Routing(CIDR)

- CIDR is a slash notation of subnet mask. CIDR tells us number of on bits in a network address.

subnet part ← → host part

11001000 00010111 00010000 00000000

200.23.16.0/23

- A single IP address can be used to designate many unique IP addresses with CIDR.
- A CIDR IP address looks like a normal IP address except that it ends with a slash followed by a number, called the IP network prefix.
- CIDR addresses reduce the size of routing tables and make more IP addresses available within organizations.

Subnetting

- Subnetting take places when we extend the default subnet mask.
- We cannot perform subnetting with default subnet mask and every classes have default subnet mask.
- Now find the host bits borrowed to create subnets and convert them in decimal.
- For example find the subnet mask of address 188.25.45.48/20 ?
  1. Class B, Default Subnet mask: 255.255.0.0
  2. Borrowed 4 bit from host part so mask is now:

     11111111 11111111 1111 0000 00000000

     255 255 240 0

How many subnets from given subnet mask?

- To calculate the number of subnets provided by given subnet mask we use 2^N , where N = number of bits borrowed from host bits to create subnets.
- For example in 192.168.1.0/27, N is 3.
- By looking at address we can determined that this address is belong to class C and default subnet mask 255.255.255.0 [/24 in CIDR].
- In given address we borrowed 27 - 24 = 3 host bits to create subnets.
- Now $2^3 = 8$, so our answer is 8.

What are the valid subnets?

- Calculating valid subnet is two steps process.
- First calculate total subnet by using formula 2^N.
- In second step find the block size and count from zero in block until subnet mask value.
- For example calculate the valid subnets for 192.168.1.0/26
  1. Borrowed host bits are 2 [26-24]
  2. Total subnets are $2^2 = 4$
  3. Subnet mask would be 255.255.255.192
  4. Block size would be 256-192 = 64
  5. Start counting from zero at blocks of 64, so our valid subnets would be 0,64,128,192

What are the total hosts?

- Total hosts are the hosts available per subnet
- To calculate total hosts use formula 2^H = Total hosts
- H is the number of host bits
- For example in address 192.168.1.0/26
- We have 32 - 26
  1. [Total bits in IP address - Bits consumed by network address] = 6
  2. Total hosts per subnet would be $2^6 = 64$

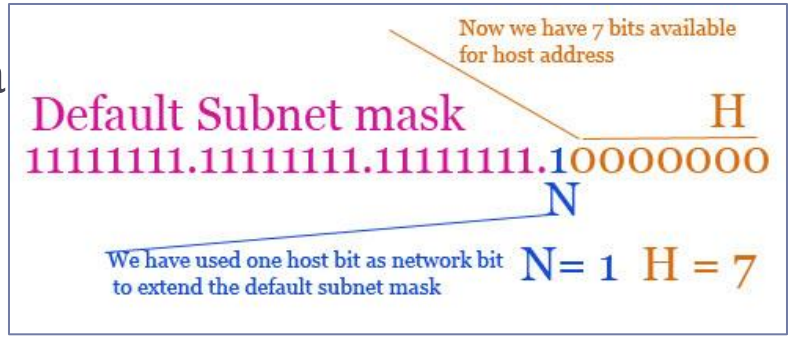
Network Prefixes

- For Class C, Default subnet mask of class C is 255.255.255.0
- CIDR notation of class C is /24, which means 24 bits from IP address are already consumed by network portion.
- We have 8 host bits remain.
- Subnetting moves from left to right. So Class C subnet mask can only be are following.

CIDR	Decimal	Binary
/25	128	10000000
/26	192	11000000
/27	224	11100000
/28	240	11110000
/29	248	11111000
/30	252	11111100

Network Prefixes- Example

- **/25**
  - CIDR /25 has subnet mask 255.255.255.128 and 128 is 10000000 in binary.
  - We used one host bit in network a



- N = 1 [Number of host bit]
- H = 7 [Remaining host bits]
- Total subnets (2^N) : 2^1 = 2
- Block size (256 - subnet mask) :- 256 - 128 = 128
- Valid subnets (Count blocks from 0) :- 0, 128
- Total hosts (2^H) :- 2^7 = 128
- Valid hosts per subnet (Total host - 2) :- 128 - 2 = 126

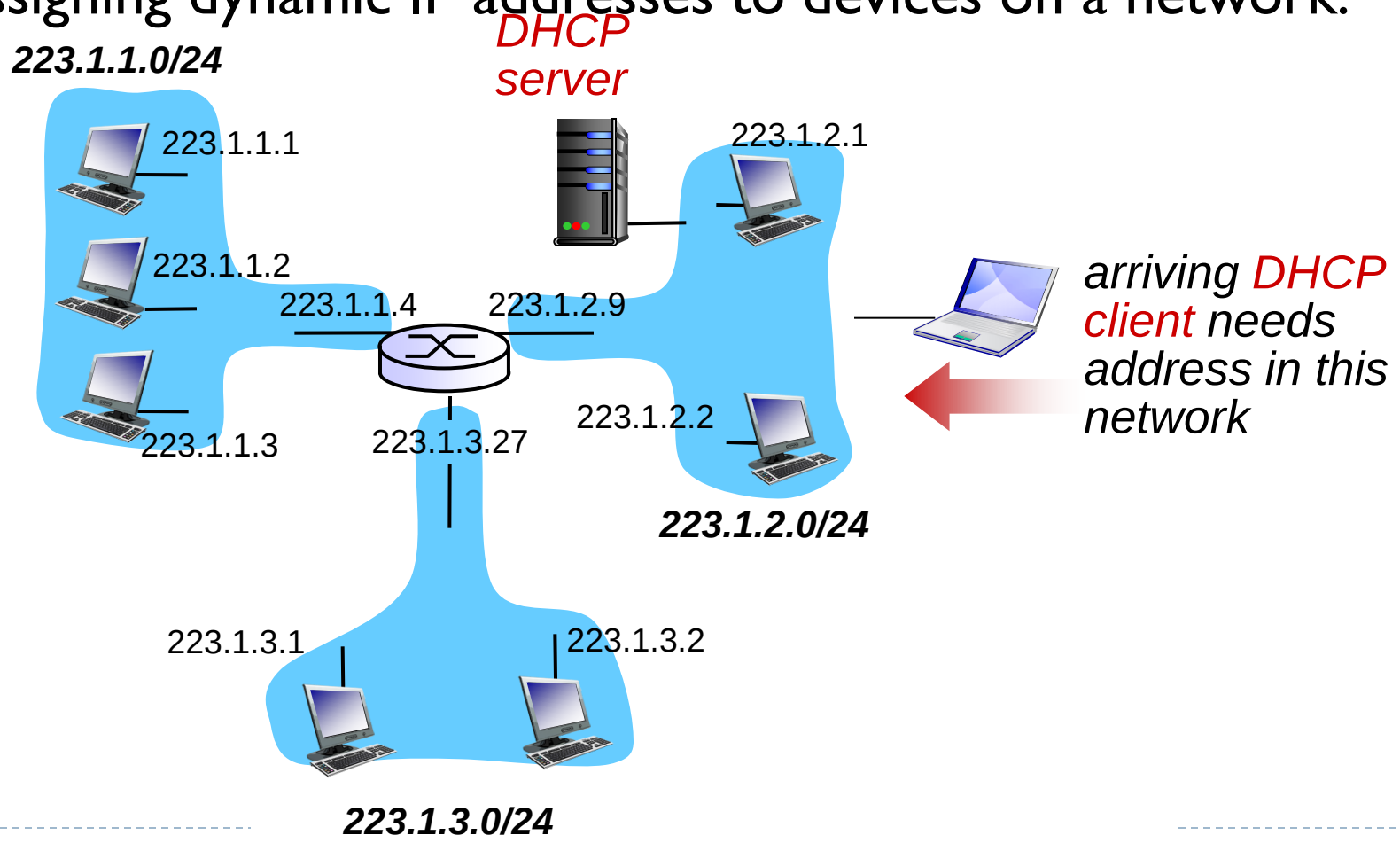
Quiz

Q) Allocate 30, 24, 25 and 12 IP addresses to four department with minimum wastage. Specify range of IP address, broadcast address, network address and subnet mask for each department from address pool 202.77.19.0/24.

Hint: *Use Variable Length Subnet Mask (VLSM)*

Dynamic Host Configuration Protocol - DHCP

- Dynamic Host Configuration Protocol is a protocol for assigning dynamic IP addresses to devices on a network.



DHCP – Cont…

- With dynamic addressing, a device can have a different IP address every time it connects to the network.
- In some systems, the device's IP address can even change while it is still connected.
- It allows reuse of addresses (only hold address while connected "on").
- It also support mobile users who want to join network.

Routing

- Whenever a device has multiple paths to reach the destination, it always selects one path. This process is called routing.
- It is done by router.
- A router is configured with some default route.
- The default route forwards a packet if no route is found for specific destination.
- If multiple paths exists to reach same destination, decision is based on : hop count, bandwidth, metric, prefix length and delay.

Static and Dynamic Routing

- Static routing is when you statically configure a router to send traffic to particular destinations in predetermined directions. It provides default route.

- Dynamic routing is when you use a routing protocol to figure out the best possible route for the traffic. It can provide the best route.

Routing Table

- Routing table is a set of rules in a table format which is used to determine where data packets traveling over an IP network will be directed.
- It contains all the information necessary to forward a packet along the best path toward its destination.
- A basic routing table includes following information:
  - Destination IP address
  - Next hop IP address
  - Outgoing network interface used
  - Cost metric to each available route
  - Routes
- Routing table can be maintained manually or dynamically.
- Dynamic routing tables allow devices to respond to device failures and network congestion.

Routing Protocols

- Routing Information Protocol (RIP)
- Open Shortest Path First (OSPF)
- Border Gateway Protocol (BGP)
- Unicast Routing
- Multicast Routing

Routing Information Protocol (RIP)

- RIP is a protocol that defines a way for routers, which connect networks using IP, to share information about how to route traffic among networks.
- Each router maintains a routing table which consist of a list of all destinations it knows how to reach and the distance to that destination.
- It uses distance vector algorithm to decide the route of packet to its destination.
- If it receives update on a route with shorter path, it will update its routing table with length and next hop address of the shorter path.
- If new route has longer path, it waits through a hold-down period and only update the table if the new route is stable.
- It follows a state of convergence. Each router sends its entire routing table to its closest neighbors every 30 seconds until all RIP hosts within the network have same knowledge of routing paths.
- It can know about router crash and if router stops sending update for six successive cycle, it will be dropped from the route.
- It uses modified hop count to determine network distance.

Open Shortest Path First (OSPF)

- It is a protocol for routers connecting networks using IP, used to find the best path for packets as they pass through a set of connected networks.
- A router which detects a change to a routing table immediately multicasts the information to all other OSPF hosts.
- It only sends the part that has changed. This helps to minimize convergence time.
- It accounts router hop as well as other network information like cost metric to determine best path.
- It has RIP support built in for compatibility with older networks using RIP.

Border Gateway Protocol (BGP)

- It is a protocol that manages how packets are routed across the Internet through the exchange of routing and reachability information between edge routers.
- It makes routing decision based on path, rules or network policies configured by a network administrator.
- Each BGP router maintains a standard routing table used to direct packets in transit.
- The table is used in conjunction with separate routing table called routing information base (RIB).
- It is based on TCP/IP and uses client-server topology.

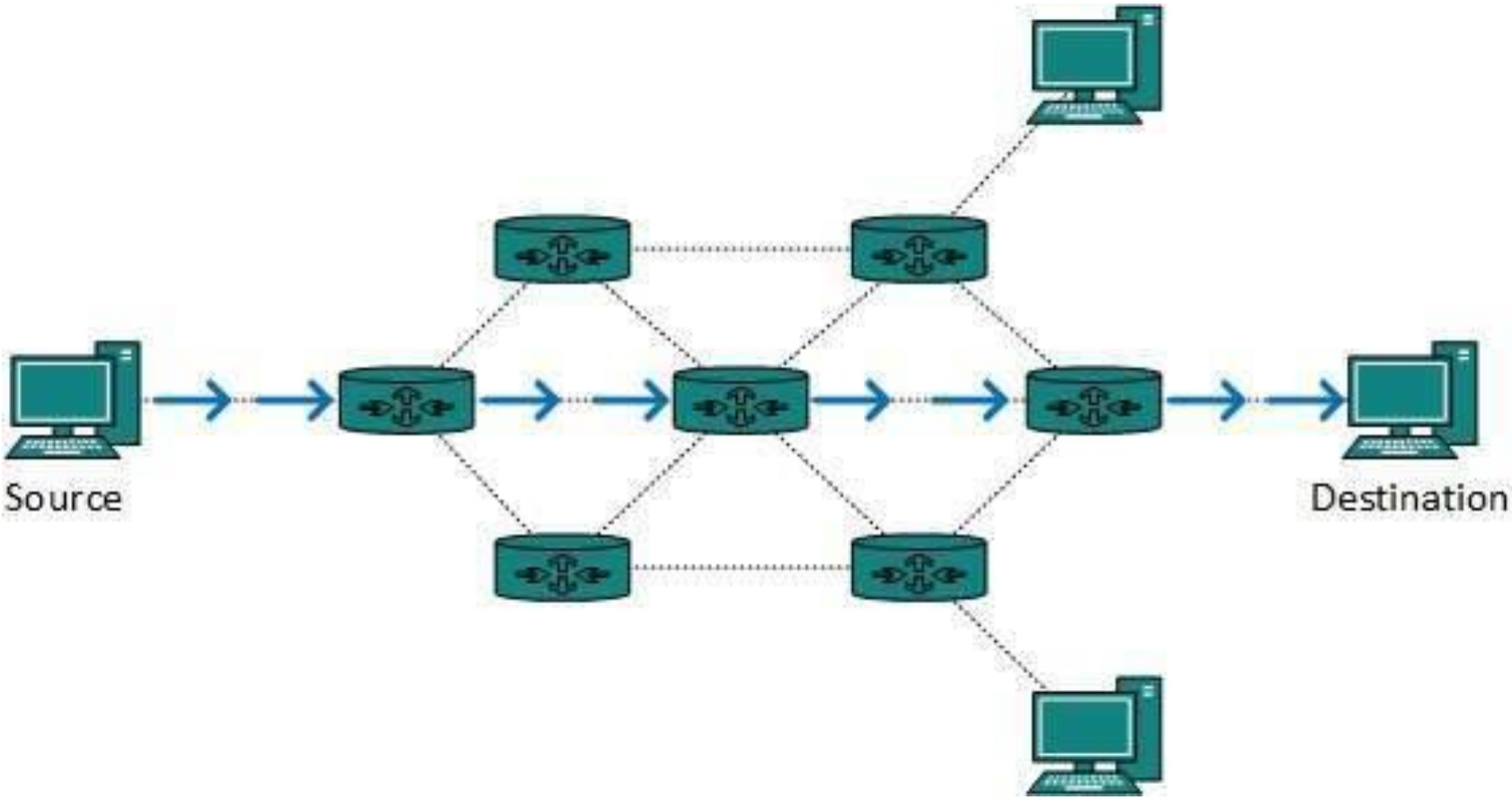
Comparison between RIP OSPF and BGP

RIP	OSPF	BGP
RIP is intra domain routing protocol used with in the autonomous system	OSPF is also intra domain routing protocol used with in the autonomous system	It is inter domain routing protocol used between the autonomous system
RIP is used for Small networks with maximum number of hops 16	OSPF is used in large autonomous system with no limitation	The BGP protocol is used for very large-scale networks
RIP uses Distance Vector	OSPF uses Link State	BGP uses Path Vector
RIP send entire routing update to all directly connected interface	OSPF send multicast Hello packet to the neighbours, to create session	BGP send Open packet to the neighbours to create session
RIP use Bellman ford Algorithm	OSPF use Dijikstra Algorithm	BGP use Path-Vector Routing

Unicast Routing

- Unicast routing is the process of routing unicast data (data sent with specified destination) over the Internet.
- The destination is already known.
- The router just has to look up the routing table and forward packet to next hop towards destination.
- For example: **Distance vector routing protocol and link state routing protocol.**
- **Read more about these protocols**

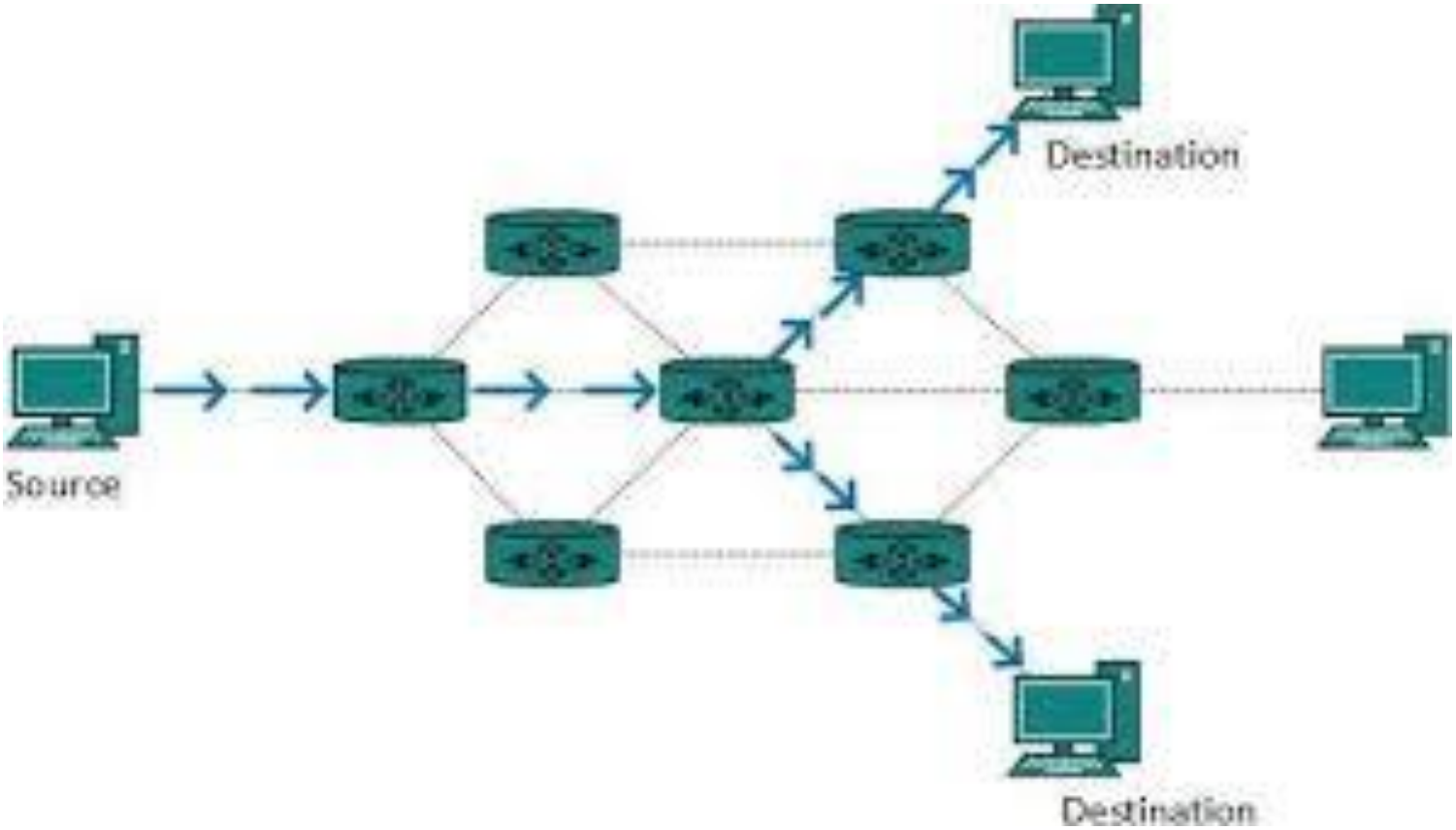
Unicast Routing



Multicast Routing

- Multicast routing is a type of broadcast routing in which the data is sent to only nodes which wants to receive the packets.
- It uses spanning tree protocol to avoid looping.
- It also uses reverse path forwarding technique.
- For example: Multicast OSPF, Core based tree, etc.

Multicast Routing



Read More ;

- **Broadcast Routing**
- **Congestion Control Algorithms**

Routing Algorithms

- Shortest Path Algorithm
- Flooding Algorithm
- Distance vector Routing
- Link State Routing

Shortest Path Algorithm

- Shortest path can be calculated only for weighted graphs.
- The edge connecting two vertices can be assigned a non-negative real number called weight of edge.
- The algorithm is:
  - Initialize array smallestWeight so that smallestWeight[u] = weights[vertex, u]
  - Set smallestWeight[vertex] = 0.
  - Find vertex v, that is closed to vertex for which shortest path has not been determined.
  - Mark v as next vertex for which smallest weight is found.
  - For each vertex w in G, such that shortest path from vertex to w has not been determined and edge (v, w) exists, if weight[u, w] < current weight, update weight of w to weight of v + weight of edge (v, w).

Shortest Path Algorithm

```
Example:
                                B
                            /       \
                        1             2
                      /                 \
Source(A) -- ---- 5 --- (D)Destination
                      \                 /
                        2             1
                          \         /
                                C


Edge        Cost          Path
-------------------------------------------
B                1             A-B
C                2             A-C
D                5             A-D
(Choose path A-B)


Edge        Cost          Path
-------------------------------------------
B                1             A-B
C                2             A-C
D                3             A-B-D
(A-B-D < A-D, So, A-B-D path is selected.)
```

Flooding Algorithm

- It is the static routing algorithm.
- Every incoming packet is sent on all outgoing lines except the line on which it has arrived.
- It generates a large number of duplicate packets on the network.

Distance Vector Routing

- Each node constructs a one dimensional array containing distances to all other nodes and distributes that vector to its immediate neighbors.
- The starting assumption is each node knows the cost of the link to directly connected neighbors only.
- A link that is down is assigned an infinite cost.
- Every node sends a message to its direct neighbors containing its personal list of distance.
- If any recipient finds the sender has path shorter than the one they know, they update the new path length.
- It should know which node tell them about the path they use.
- Each node maintains forwarding table.

Link State Routing

- Each router knows about its direct neighbors.
- Each router constructs a link state packet (LSP), which consists of:
  - ID of node that create LSP.
  - A list of direct neighbors and cost of link to each one.
  - Sequence number
  - A time to line (TTL) for the packet.
- Link state flooding is done.
- Each router stores most recently generated LSP from each other router.
- Shortest path route to each destination is computed.

Differences: LS and DV Routing Algorithm

Distance Vector Protocol	Link State Protocol
Entire routing table is sent as an update	Updates are incremental & entire routing table is not sent as update
Distance vector protocol send periodic update at every 30 or 90 second	Updates are triggered not periodic
Update are broadcasted	Updates are multicasted
Updates are sent to directly connected neighbour only	Update are sent to entire network & to just directly connected neighbour
Routers don't have end to end visibility of entire network.	Routers have visibility of entire network of that area only.
It is prone to routing loops	No routing loops

Further Reading

- Address Resolution Protocol (ARP)
- Reverse ARP (RARP)
- Internet Control Message Protocol (ICMP)
- Internet Protocol (IP)

Difference between IPv4 & IPv6

IPv4	IPv6
✓ 32 bit length	✓ 128 bit length
✓ Fragmentation is done by sender and forwarding routers	✓ Fragmentation is done only by sender
✓ No packet flow identification	✓ Packet flow identification is available within the IPv6 header using the Flow Label field
✓ Checksum field in header	✓ No checksum field in header
✓ Options fields are available in header	✓ No option fields, but Extension headers are available
✓ Address Resolution Protocol (ARP)is available to map IPv4 addresses to MAC addresses	✓ Address Resolution Protocol (ARP) is replaced with Neighbor Discovery Protocol
✓ Broadcast messages are available	✓ Broadcast messages are not available
✓ Static IP addresses or DHCP is required to configure IP addresses	✓ Auto-configuration of addresses is available

Test Quiz

- The network layer concerns with
  a) bits
  b) frames
  c) packets
  d) none of the mentioned

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Test Quiz

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  b) inter-networking
  c) congestion control
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  a) network address
  b) host address
  c) both network address & host address
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  a) ethernet
  b) internet protocol
  c) hypertext transfer protocol
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  a) error and diagnostic functions
  b) addressing
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